CIE xyZ color space

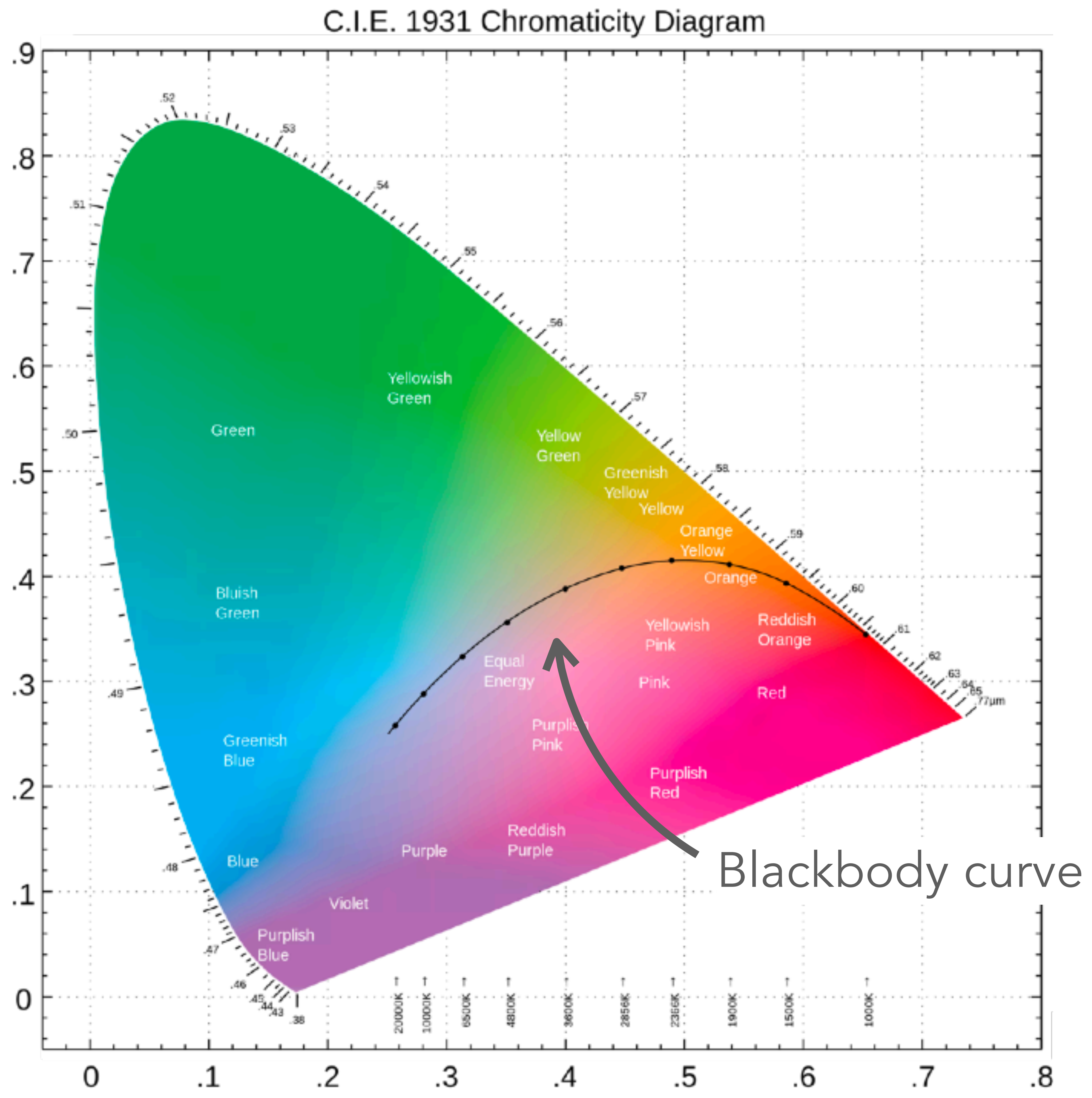
xyZ: a **chromaticity** model that isolates color (chromaticity) from luminance

$$x = \frac{X}{X+Y+Z}$$

$$y = \frac{Y}{X+Y+Z}$$

$$z = \frac{Z}{X+Y+Z} = 1 - x - y$$

Chromaticity diagram: project onto 2D



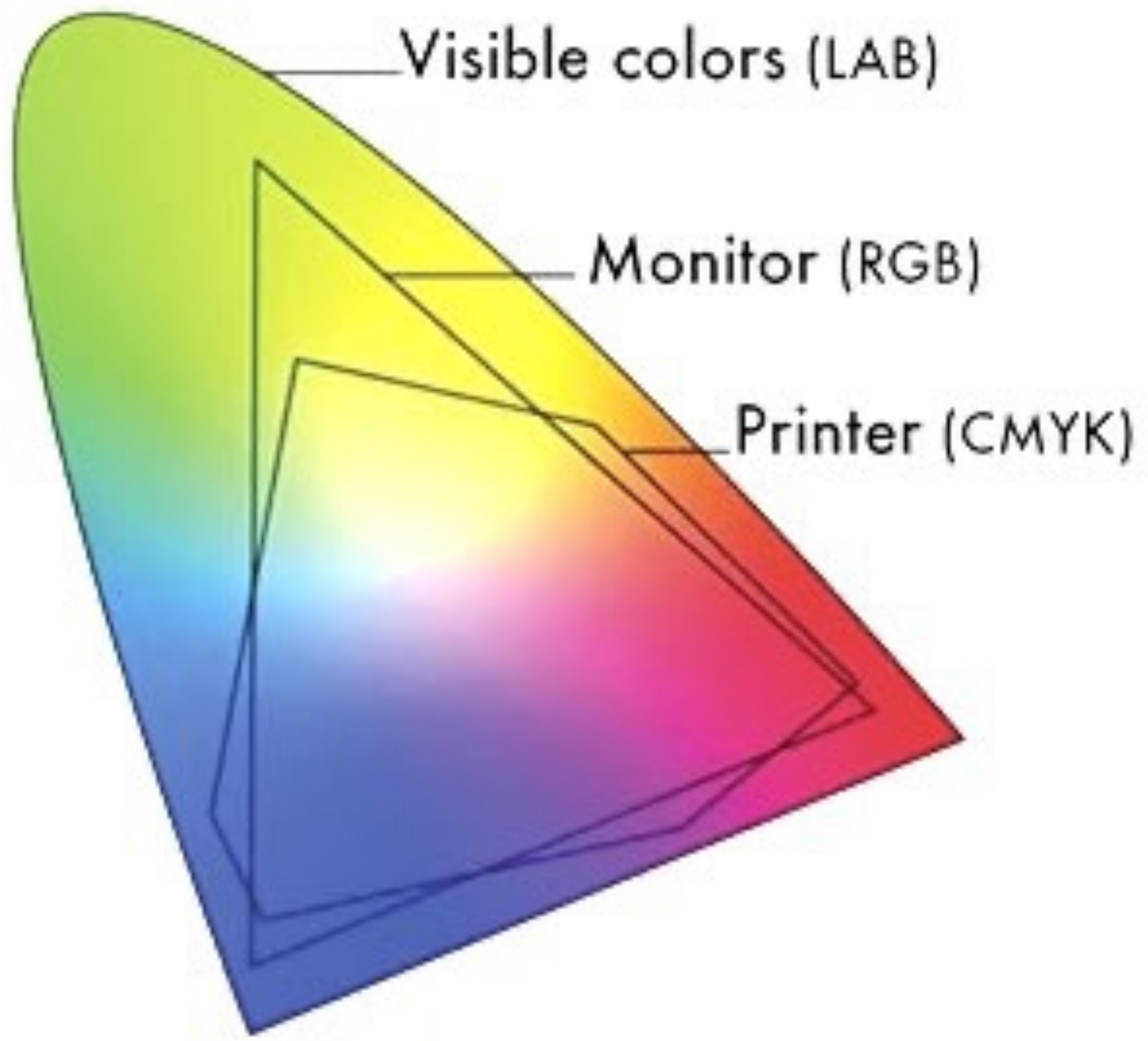
Color spaces

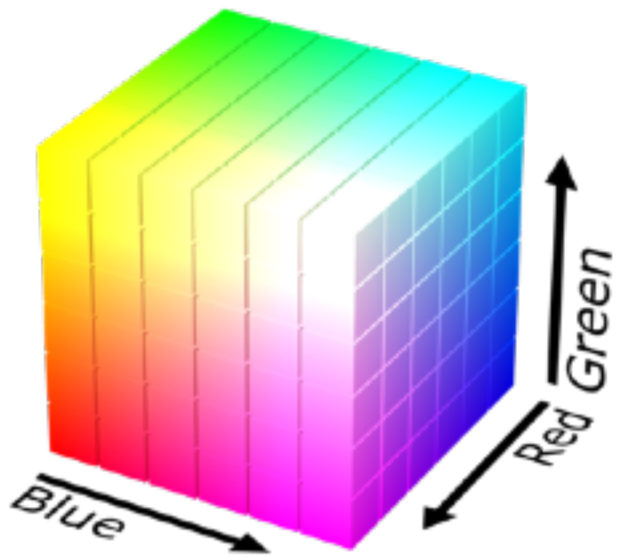
CIE La*b*: perceptually uniform, changes in the values correspond more closely to perceived differences in color. Scaling of axes to represent "color distance"

RGB: digital displays and imaging

CMYK: printers; **HSL**, **HSV**, **HSK**, etc…

conversion between color spaces:
https://medium.com/@alakhsharmacs/rgb-to-lab-color-space-conversion-formulas-insights-and-applications-7fd6efa519dd





https://pagesthemagazine.com/color-me-confident/